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Stability of finite difference schemes

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Last time

- The advection equation $\partial_t u + v\,\partial_x u = 0$ carries a profile along at speed v: $u = u_0(x - vt)$.
- Characteristics: the curves along which a first-order PDE is an ODE. Where they converge the profile steepens; where they cross, a shock forms.
- The value at (x, t) depends on the initial data in a definite place, the domain of dependence. A scheme had better look there.
- The first scheme we tried, FTCS, blew up.

Today: exactly why it blew up, a test that tells in advance, and three ways to fix it, each with a price.

Part I · Why FTCS blows up

Follow one wave through one step

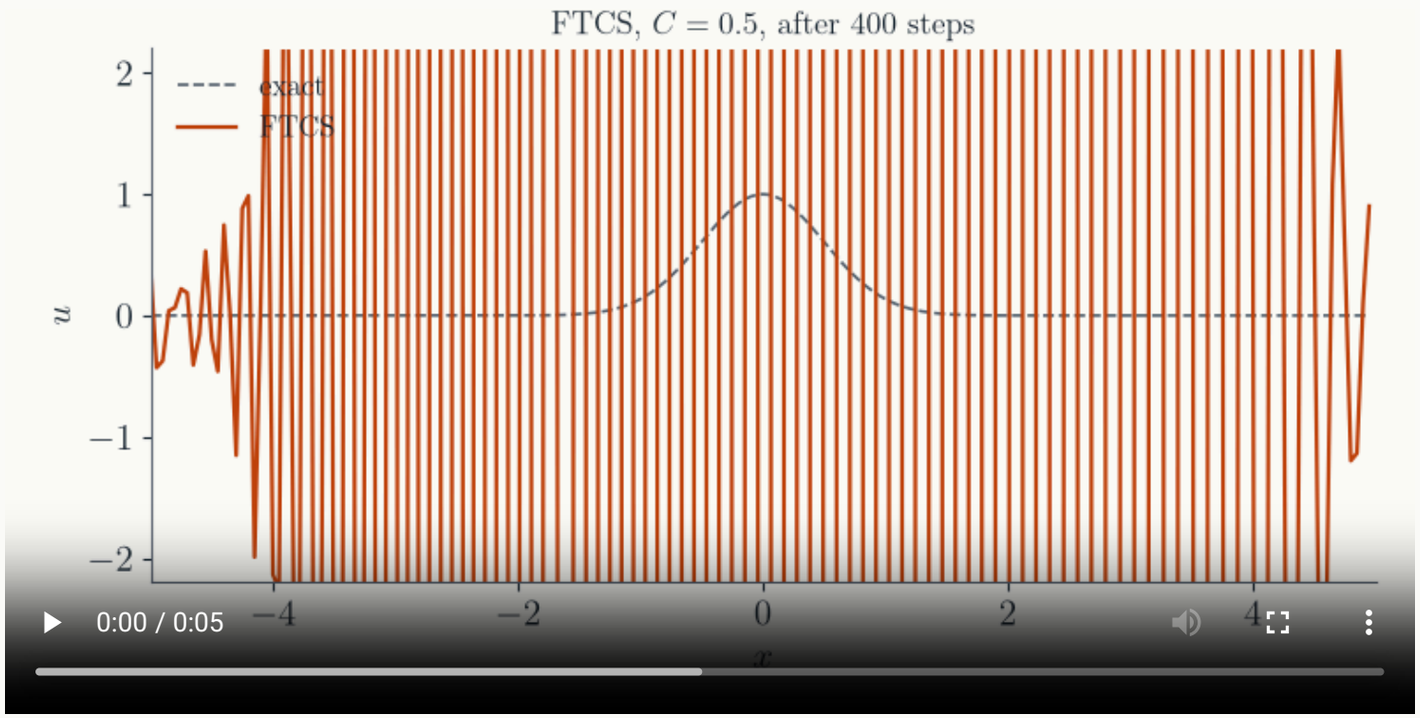
The scheme again

Points $x_j = j\,\Delta x$, times $t_n = n\,\Delta t$, unknowns u_j^n. Forward difference in time, centered in space:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + v\,\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0 \quad\Longrightarrow\quad u_j^{n+1} = u_j^n - \frac{C}{2}\left(u_{j+1}^n - u_{j-1}^n\right), \quad C = \frac{v\Delta t}{\Delta x}.$$

- C, the Courant number, is how many grid spacings the flow moves in one step. It is the only parameter of the scheme.
- Euler's method on the system of ODEs $du_j/dt = -v\,(u_{j+1} - u_{j-1})/2\Delta x$, one per grid point. Both halves are things we trust.

FTCS fails



- $C = 0.5$, 200 points, periodic box. The Gaussian survives a while, then a grid-scale oscillation appears, and by 400 steps it is at 2×10^3.
- A smaller Δt delays it and does not cure it. To see why, stop looking at the Gaussian and look at one wave.

Stability Analysis

Unfortunately, the FTCS scheme is not very stable. To understand what is going on, we can use the *von Neumann stability analysis*.

Let's assume that a wave exists on our grid that takes the form:

$$u(x,0) = e^{ikx}, \quad u_j^0 = e^{ikx_j}$$

where k is a constant. What happens to this wave after one time step?

Stability Analysis

After one step of the FTCS scheme, we have:

$$
\begin{aligned}
u_j^1 &= u_j^0 - v\frac{\Delta t}{2\Delta x}(u_{j+1}^0 - u_{j-1}^0) \\
&= u_j^0 - v\frac{\Delta t}{2\Delta x}(e^{ik\Delta x} - e^{-ik\Delta x})u_j^0 \\
&= \left[1 - i\frac{v\Delta t}{\Delta x}\sin(k\Delta x)\right] u_j^0 \\
&= \xi u_j^0
\end{aligned}
$$

After many steps, the solution will look like $u_j^n = \xi^n u_j^0$.

Stability Analysis

We want our solution to not grow to infinity after some time. Therefore, the stability condition is simply $|\xi| \leq 1$. ξ is called the "amplification factor" of the numerical scheme. Note that ξ is often complex, therefore $|\xi|$ means its modulus.

From our previous derivation, $|\xi|$ for the FTCS scheme is:

$$|\xi| = \left|1 - i\frac{v\Delta t}{\Delta x}\sin(k\Delta x)\right| = \sqrt{1 + \left(\frac{v\Delta t}{\Delta x}\right)^2 \sin^2(k\Delta x)}$$

This is larger than 1 for any k, Δx, and Δt. Therefore we call the FTCS scheme *unconditionally unstable*.

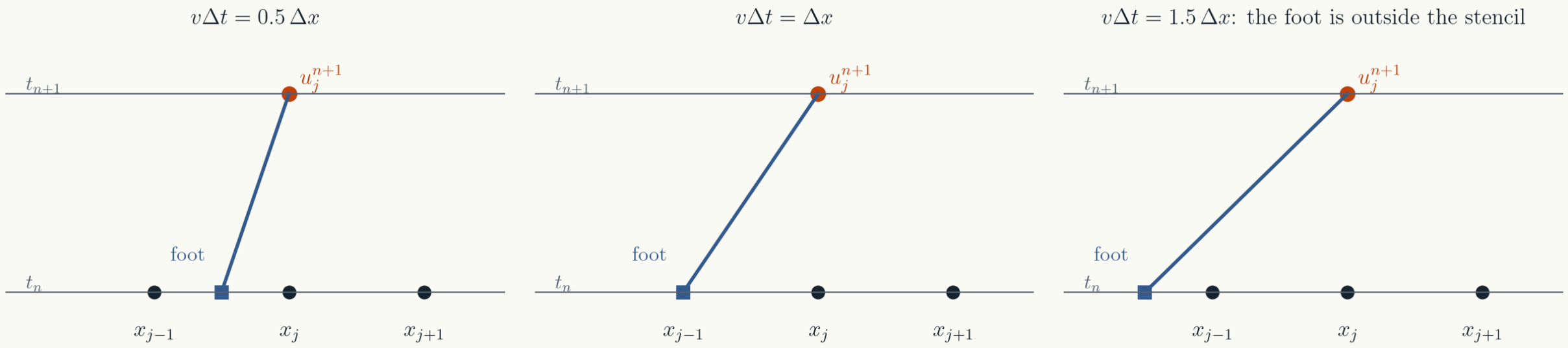
Part II · Fix 1: take information from upwind

Ask the characteristic

Ask the characteristic

The exact solution says where the new value comes from: go down the characteristic for one step,

$$u(x_j, t_{n+1}) = u(x_j - v\Delta t,\ t_n).$$



- The foot, $x_j - v\Delta t$, is not a grid point. Read the old solution there by linear interpolation between the two grid points around it, x_{j-1} and x_j.

The upwind scheme

The foot lies a fraction C of the way from x_j back to x_{j-1}. Linear interpolation gives

$$u_j^{n+1} = (1 - C)\, u_j^n + C\, u_{j-1}^n \qquad \Longleftrightarrow \qquad \frac{u_j^{n+1} - u_j^n}{\Delta t} + v\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0.$$

- A one-sided difference, taken from the side the flow comes from. For $v < 0$ use $u_{j+1} - u_j$ instead.
- Same cost as FTCS, and it only ever reads information from where the characteristic says it is.

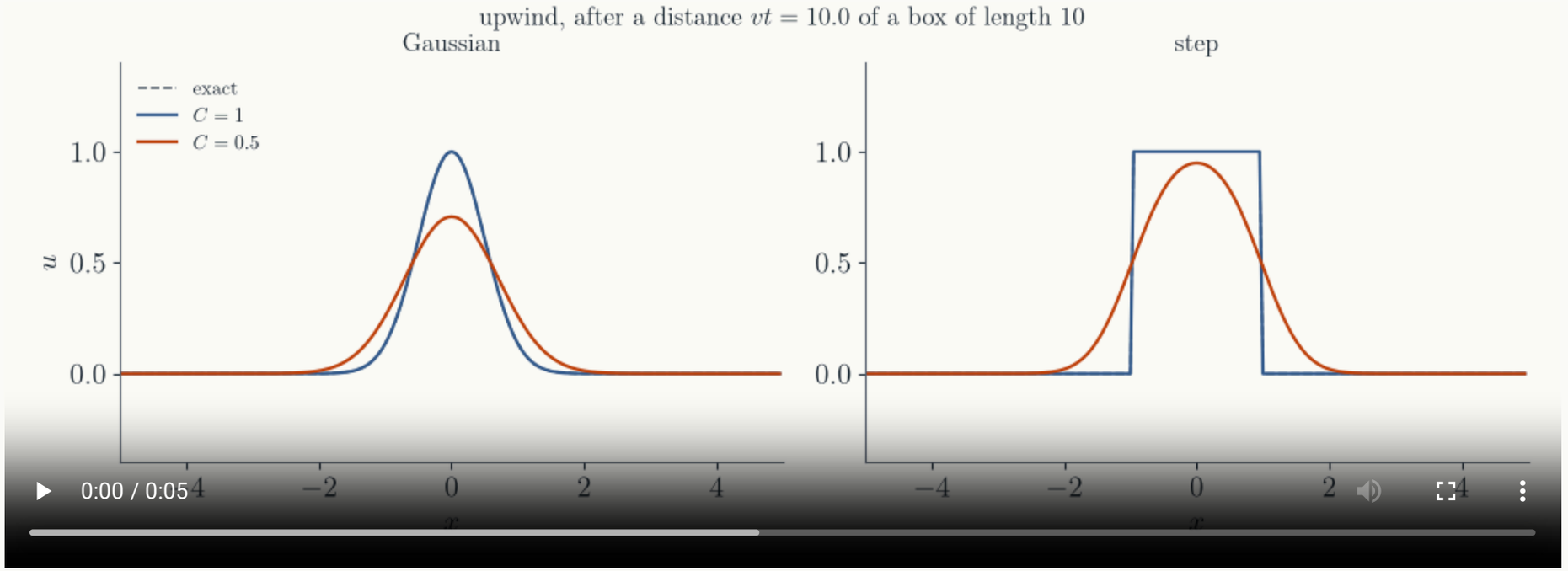
Its amplification factor

Same substitution, $u_j^n = \xi^n e^{ikx_j}$:

$$\xi = 1 - C + C\,e^{-ik\Delta x}, \qquad |\xi|^2 = 1 - 2C(1-C)\left(1 - \cos k\Delta x\right).$$

- $1 - \cos k\Delta x \geq 0$, so $|\xi| \leq 1$ for every wave exactly when $C(1-C) \geq 0$: that is $0 \leq C \leq 1$.
- This is the *CFL condition* (Courant, Friedrichs, Lewy, 1928): $v\,\Delta t \leq \Delta x$. In one step, the flow may not travel farther than one cell.
- In the stencil picture: the foot of the characteristic must lie between the two points the scheme reads. For $C > 1$ it lies beyond x_{j-1}, the interpolation becomes an extrapolation with a negative weight $1 - C$, and the errors grow every step.
- At $C = 1, \xi = e^{-ik\Delta x}$: modulus one and the exact phase. The foot lands on x_{j-1} and the scheme is the exact shift.

Upwind in action



- One crossing of the box. At $C = 1$ both profiles return exactly. At $C = 0.5$ the Gaussian peak has fallen to 0.71 and the step's corners are gone.
- Stable, and smeared. The next slide says by how much, and why.

The price: numerical diffusion

Expand the scheme in Taylor series about (x_j, t_n) and keep one more order than the equation it was meant to solve. What the upwind scheme actually solves is

$$\frac{\partial u}{\partial t} + v\,\frac{\partial u}{\partial x} = D\,\frac{\partial^2 u}{\partial x^2} + \dots, \qquad D = \tfrac{1}{2}\,v\,\Delta x\,(1 - C).$$

- This is the *modified equation*: advection plus a diffusion term that the scheme added, with a coefficient set by the grid. It vanishes at $C = 1$ and is largest for small steps: at $C = 0.5$, $D = 0.25\,v\Delta x$.
- Diffusion is what averaging does. Each step replaces u_j by a weighted mean of itself and its upwind neighbour, and means are smoother than what they average.

The scheme is first order: the error term is proportional to Δx. Halve the grid and the smearing halves.

Part III · Fix 2: add diffusion on purpose

Lax-Friedrichs

The Lax idea

Keep the centered difference, but replace the u_j^n that the step starts from by the average of its neighbours:

$$u_j^{n+1} = \frac{1}{2}\left(u_{j+1}^n + u_{j-1}^n\right) - \frac{C}{2}\left(u_{j+1}^n - u_{j-1}^n\right).$$

- Amplification factor: $\xi = \cos k\Delta x - i\,C \sin k\Delta x$, so $|\xi|^2 = \cos^2 k\Delta x + C^2 \sin^2 k\Delta x = 1 - (1 - C^2)\sin^2 k\Delta x$.
- $|\xi| \le 1$ for every wave exactly when $C \le 1$. Stable, with the same condition as upwind, and it never had to ask which way the wind blows.

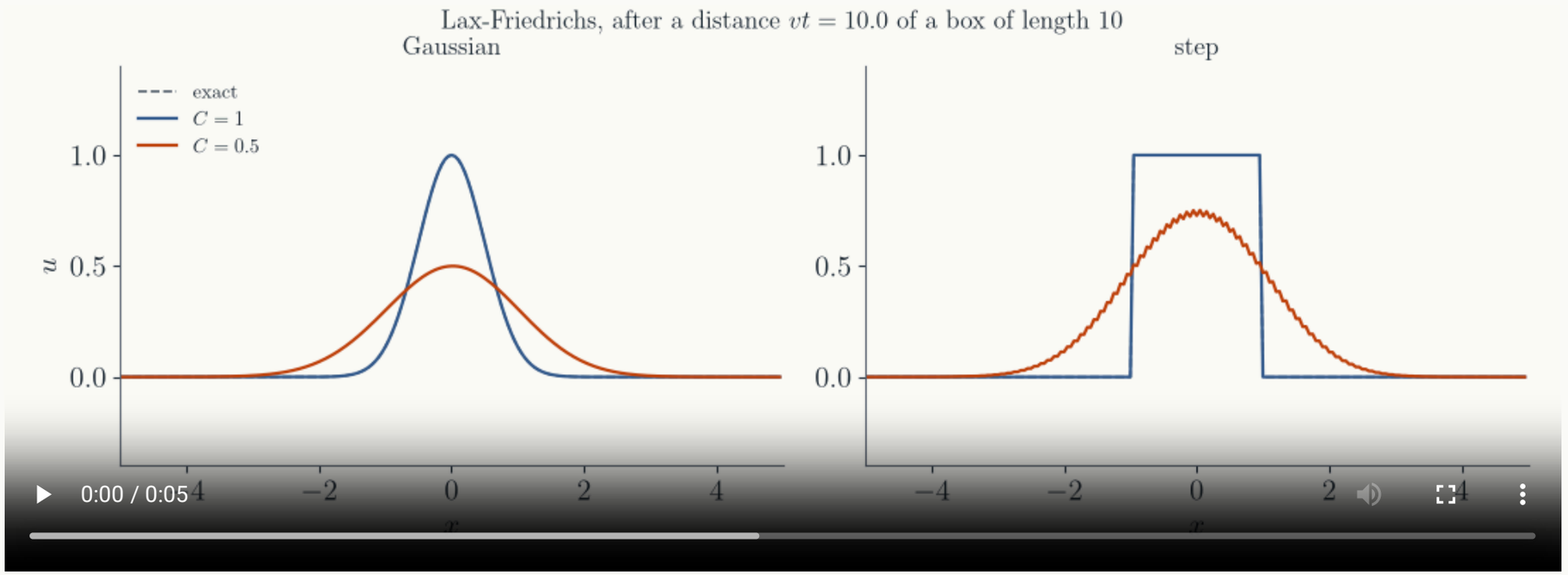
What was added

Subtract u_j^n from both sides and the average becomes a second difference:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + v\,\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = \frac{\Delta x^2}{2\Delta t}\,\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}.$$

- FTCS on the left, a diffusion term on the right with $D = \Delta x^2/2\Delta t = v\Delta x/2C$. Diffusion damps the short waves that FTCS amplified; enough of it, and nothing grows.
- The price is the same as upwind's, and larger: at $C = 0.5$, $D = 0.75\,v\Delta x$ against 0.25. The Gaussian peak after one crossing is 0.50, against 0.71.
- One wave escapes: at $k\Delta x = \pi$, the two-cell checkerboard, $|\xi| = 1$ at every C. The average of the neighbours of a checkerboard is the checkerboard. It shows as fine ripples on the step.

Lax-Friedrichs in action



- Exact at $C = 1$, like upwind, and for the same reason: the scheme becomes a shift.
- At $C = 0.5$ it smears three times faster than upwind, and the checkerboard ripples ride on the step.

Part IV · Fix 3: be second order

Lax-Wendroff

One more term of Taylor

Euler in time keeps one term of the Taylor series. Keep two, and use the equation itself to turn time derivatives into space derivatives:

$$u^{n+1} = u + \Delta t\,\frac{\partial u}{\partial t} + \frac{\Delta t^2}{2}\,\frac{\partial^2 u}{\partial t^2}, \qquad \frac{\partial u}{\partial t} = -v\,\frac{\partial u}{\partial x}, \qquad \frac{\partial^2 u}{\partial t^2} = v^2\,\frac{\partial^2 u}{\partial x^2}.$$

Centered differences for both space derivatives give the *Lax-Wendroff* scheme:

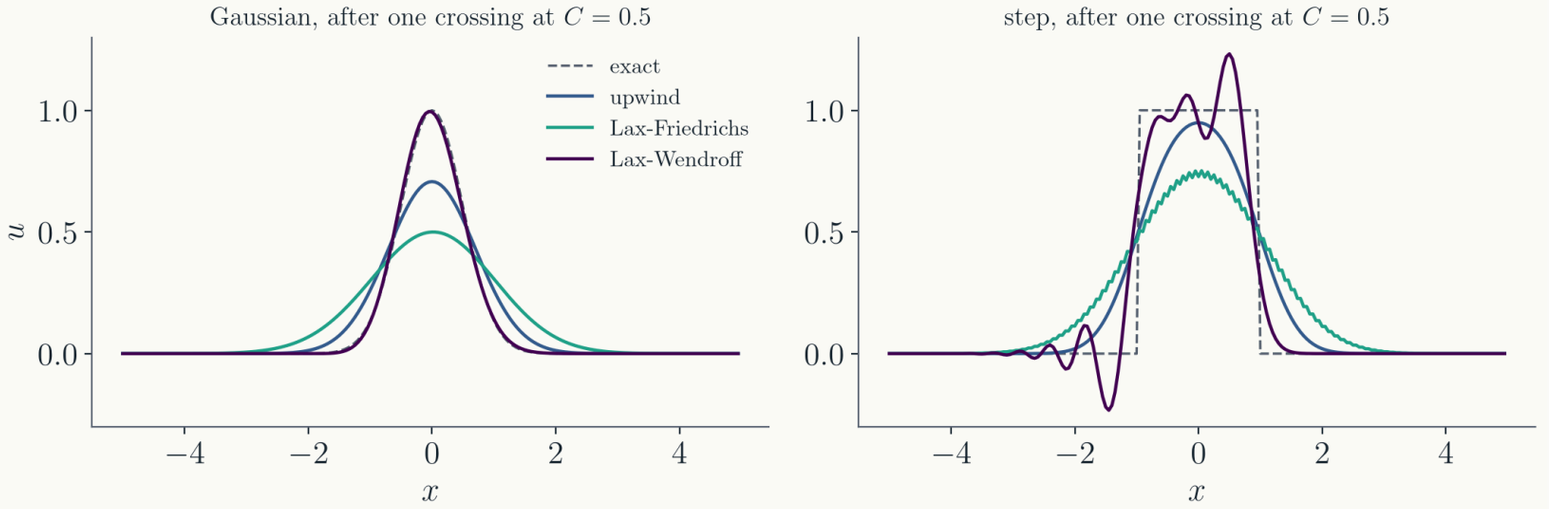
$$u_j^{n+1} = u_j^n - \frac{C}{2}\left(u_{j+1}^n - u_{j-1}^n\right) + \frac{C^2}{2}\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right).$$

FTCS plus a second difference, like Lax, but with the coefficient $C^2/2$ that the Taylor series asks for instead of the $1/2$ that Lax put in by hand. Second order in Δt and Δx.

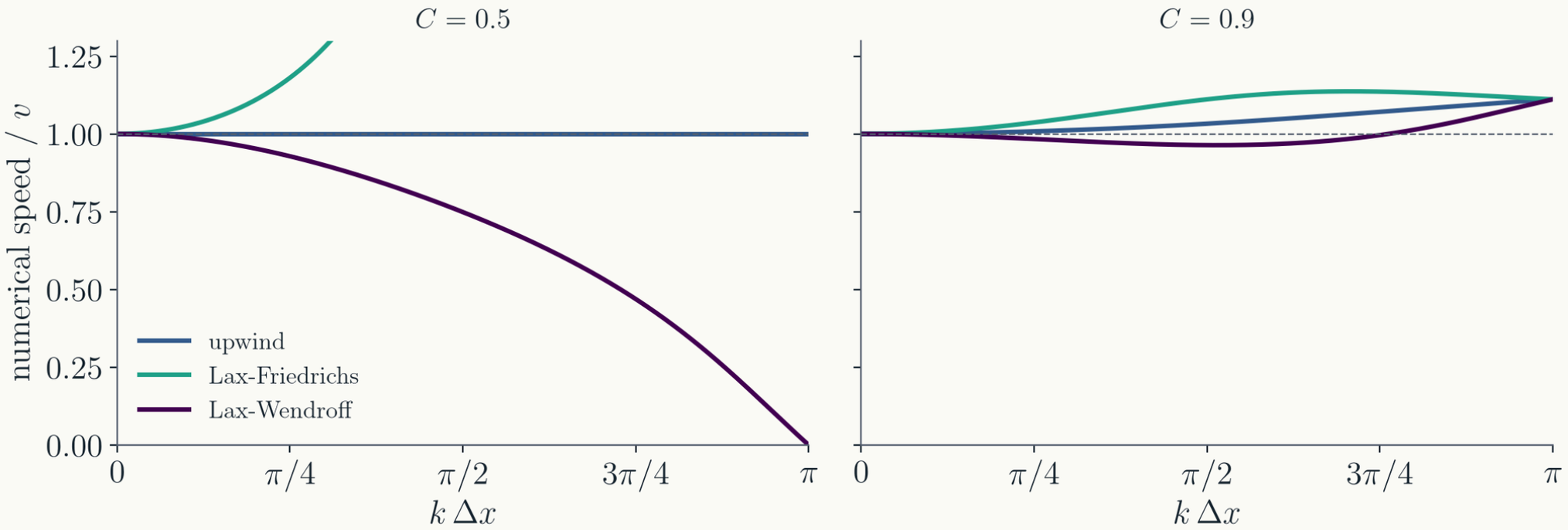
Stable, and accurate

$$\xi = 1 - i\,C \sin k\Delta x - C^2\,(1 - \cos k\Delta x)\,, \qquad |\xi|^2 = 1 - C^2\,\left(1 - C^2\right)\,(1 - \cos k\Delta x)^2.$$

- $|\xi| \le 1$ for every wave exactly when $C \le 1$. Same condition a third time.
- The damping is fourth order in $k\Delta x$ for long waves, where upwind's was second order. The Gaussian comes back from one crossing at $C = 0.5$ with its peak at 0.995: against 0.71 for upwind and 0.50 for Lax.

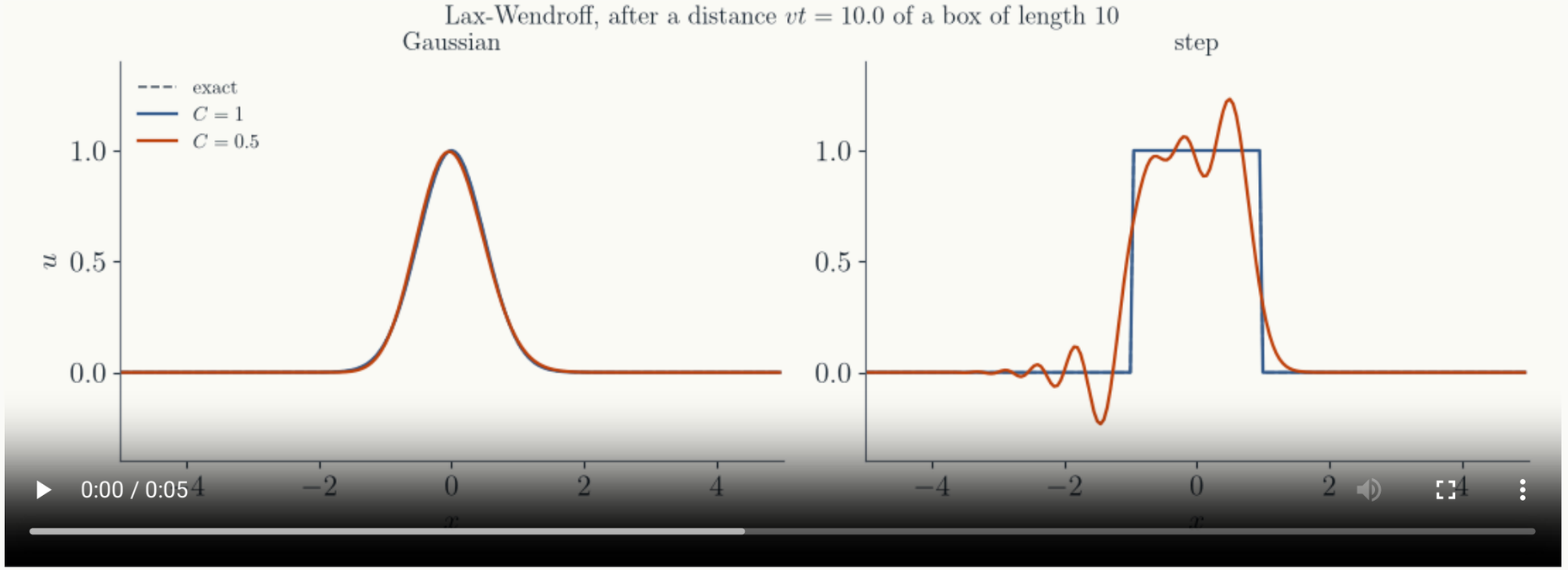


The price: dispersion



- The phase of ξ is how far the wave moved in one step. Exact: $-Ck\Delta x$. Lax-Wendroff moves short waves too slowly, $0.75\,v$ at $k\Delta x = \pi/2$ for $C = 0.5$, and the two-cell wave not at all.
- A step is made of short waves. They fall behind, and the step arrives with a train of oscillations trailing each edge: overshoot 1.23, undershoot -0.23 after one crossing. Lax-Friedrichs has the opposite error: its short waves run ahead.

Lax-Wendroff in action

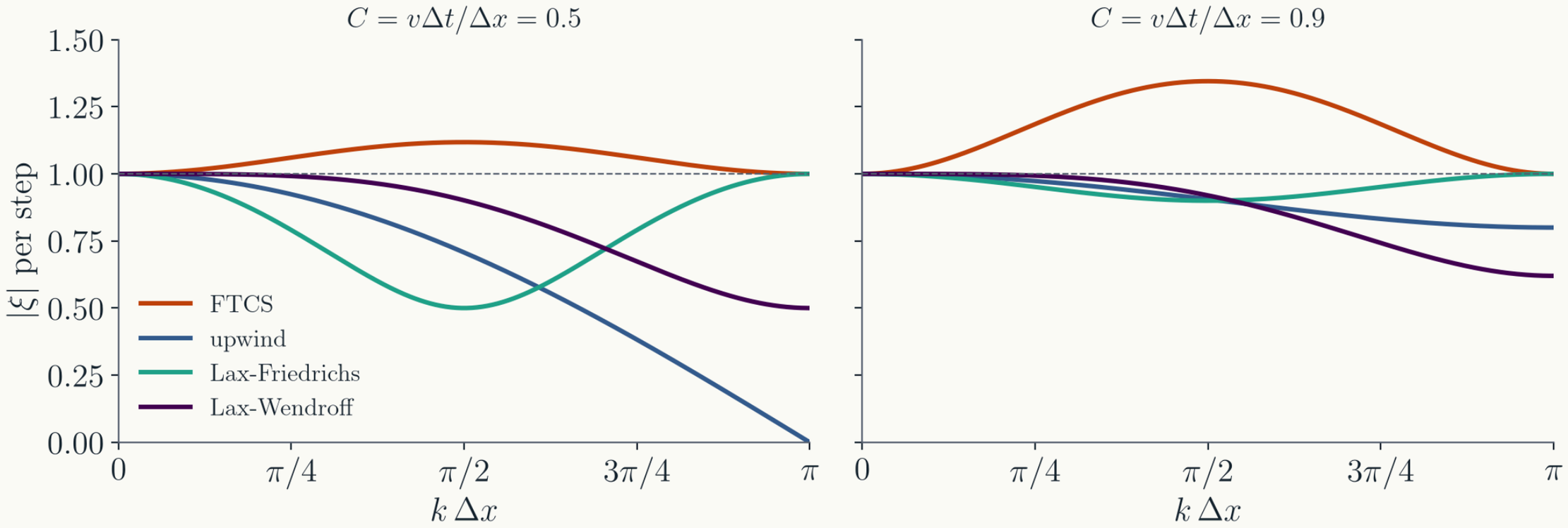


- The Gaussian is carried almost untouched, which neither of the first-order schemes could do.
- The step grows wiggles behind each edge, which neither of the first-order schemes did. Second order buys accuracy on smooth solutions and pays for it at jumps.

Part V · Choosing

Diffusion or dispersion

The four amplification factors



- The three fixes all sit at or below 1, and all touch 1 at $k\Delta x = 0$: long waves are carried faithfully by every stable scheme.
- They differ in how hard they damp the short waves. Lax-Friedrichs damps the four-cell wave to 0.5 per step at $C = 0.5$; Lax-Wendroff leaves it at 0.90. That is the diffusion difference of the crossing figure, wave by wave.

Three schemes, one table

scheme	reads	order	stable for	leading error	Gaussian peak after one crossing, $C = 0.5$	step
FTCS	$j-1, j, j+1$	1	never	growth	blows up	blows up
upwind	$j-1, j$	1	$C \le 1$	diffusion, $\frac{1}{2}v\Delta x(1-C)$	0.71	smeared
Lax-Friedrichs	$j-1, j+1$	1	$C \le 1$	diffusion, $\frac{1}{2}v\Delta x(1-C^2)/C$	0.50	smeared, ripples
Lax-Wendroff	$j-1, j, j+1$	2	$C \le 1$	dispersion, $\frac{1}{6}v\Delta x^2(1-C^2)$	0.995	wiggles

- All three stable schemes are exact at $C = 1$ and worst at small C: a smaller time step is *not* more accurate here. Run as close to the CFL limit as you dare.
- First order smears, second order wiggles. The wiggles are confined to jumps; the smearing is everywhere.

Which to use

- Smooth solutions: a second-order scheme, or higher. The dispersion error shrinks as Δx^2 and there are no jumps to excite it.
- Solutions with shocks or fronts: you need diffusion at the jump, and would rather not have it elsewhere. The modern answer is a second-order scheme that switches to first order where the solution is not smooth, a *limiter*, and a flux with just enough built-in diffusion to cover the fastest wave.

Every scheme has an error. The choice is what kind, and where.

Summary

- **Von Neumann:** put $\xi^n e^{ikx_j}$ into the scheme; it is stable if $|\xi| \leq 1$ for every k. One line per scheme, and it predicts the blow-up before any run.
- **FTCS:** $|\xi|^2 = 1 + C^2 \sin^2 k\Delta x$, unstable at every C.
- **Three fixes, one condition:** upwind, Lax-Friedrichs and Lax-Wendroff are all stable for $C \leq 1$ and exact at $C = 1$.
- **Two prices:** first-order schemes add diffusion, $D \propto v\Delta x$; the second-order scheme adds dispersion and wiggles at jumps. The modified equation says which, and how much.

Next time: the diffusion equation, where the explicit scheme is stable but the time step is tiny, and the implicit schemes that fix it. Then the wave equation on a staggered grid.